21.11.2025 (M)

SOMAIYA VIDYAVIHAR UNIVERSITY

| Semester: July 2025 –November 2025 | | |
|---|---|---|
| **Maximum Marks: 100** | **Examination: ESE Examination** | **Duration:3 Hrs.** |
| **Programme code: 01**<br>**Programme: B.Tech Computer Engineering** | **Class:**<br>**LY BTech** | **Semester: VII (SVU 2020)** |
| **Institute/School/Department: K. J. Somaiya School of Engineering** | **Name of the department:**<br>**Computer Engineering** | |
| **Course Code: 116U01E747** | **Name of the Course: DevOps** | |
| **Instructions: 1)Draw neat diagrams 2) All questions are compulsory**<br>**3) Assume suitable data wherever necessary** | | |

| Que. No. | Question | Max. Marks |
|---|---|---|
| Q1 | Solve *any* **Four** | **20** |
| i) | List any five benefits of adopting DevOps in an organization. | 5 |
| ii) | Define affinity and explain its significance in achieving high-performing teams. | 5 |
| iii) | What is the role of version control systems (VCS) in DevOps? Give suitable example. | 5 |
| iv) | Explain the purpose and use of JUnit in automated testing. | 5 |
| v) | Mention how monitoring tools like Nagios, Munin, and Ganglia support system health management. | 5 |
| vi) | Differentiate between AWS and Azure in deployment automation. | 5 |

| Que. No. | Question | Max. Marks |
|---|---|---|
| Q2 A | Solve the following | **10** |
| i) | What are the main goals of Development, Release, and Deployment Concepts in DevOps? | 5 |
| ii) | Analyze how empathy and trust contribute to the success of DevOps transformations. | 5 |
| | **OR** | |
| Q2 A | A retail company wants to deploy its application to 20 servers daily with minimal downtime.<br>(a) Which combination of deployment tools would you suggest?<br>(b) Explain how job scheduling, triggers, and pipelines would be set up in Jenkins for automated deployment.<br>(c) How would you manage rollback if a deployment fails? | 10 |
| | | |
| Q 2 B | Solve any **One** | **10** |
| i) | Your organization runs a large monolithic payroll application built 10 years ago. It has become difficult to update, and deployments take hours. The management wants to move toward a faster and more scalable architecture.<br>(a) Identify the key limitations of the monolithic scenario in this case.<br>(b) Explain how DevOps practices (CI/CD) can support this transition. | 10 |
| ii) | Describe the architecture and functioning of a source code management system. | 10 |

| Que. No. | Question | Max. Marks |
|---|---|---|
| Q3 | Solve *any* **Two** | **20** |
| i) | Analyze how scaling DevOps practices differs at the team level versus the organizational level. | 10 |
| ii) | With suitable examples, discuss how architecture decisions (monolithic vs. microservices) and source code management are interlinked in modern DevOps workflows. | 10 |
| iii) | A Jenkins pipeline triggered by a Git commit failed because of a missing configuration file.<br>(a) Analyze how Git hooks or branch protection rules could prevent such errors.<br>(b) Suggest improvements to integrate Git and Jenkins effectively in the CI/CD pipeline. | 10 |

| Que. No. | Question | Max. Marks |
|---|---|---|
| Q4 | Solve *any* **Two** | **20** |
| i) | Describe strategies for ensuring robustness and fault tolerance in automated build pipelines. | 10 |
| ii) | Explain how Docker Compose simplifies multi-container deployment. Give an example of docker-compose.yml file. | 10 |
| iii) | Describe the role of Kubernetes YAML manifests or Helm charts in automating deployment. Give example. | 10 |

| Que. No. | Question | Max. Marks |
|---|---|---|
| Q5 | Solve *any* **four** | **20** |
| i) | DevOps as a Folk Model. | 5 |
| ii) | Learning Organizations. | 5 |
| iii) | Architecture rules of thumb. | 5 |
| iv) | Triggers & Job Chaining | 5 |
| v) | Infrastructure as a Code (IaC). | 5 |
| vi) | Fault tolerance in Dockers & Kubernetes | 5 |